

G H RAISONNI COLLEGE OF ENGINEERING & MANAGEMENT

Accredited by NAAC with A+ Grade

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Department of MCA

MCA First Year, Semester-II

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Subject: 24MCAL204: DATA STRUCTURE

UNIT – I

Que 1. a) What is Data Structure? Explain its types in details. (7)

b) Define Abstract Data Type (ADT). (3)

Ques 2. a) Explain Linear Data structure with examples. (5)

b) Explain Non-Linear Data structure with example. (5)

Que 3. a) Differentiate between Static and Dynamic Memory Allocation. (5)

b) Explain Array with Example. (5)

Que 4. a) Define Array. Explain how arrays are represented in computer memory. (7)

b) Write algorithm for accessing the element of an array. (3)

Que 5. What are the operations that can be performed on arrays. Explain it with an example. (10)

Que 6. a) Write an algorithm for

- i. Algorithm to append a new element to an existing array. (1)
- ii. Algorithm to insert an element in the middle of an array. (4)
- iii. Algorithm to delete the last element of an array. (1)
- iv. Algorithm to delete an element from the middle of an array. (4)

Que 7) Given an array `int marks[]={99,67,78,56,88,90,34,85}`, calculate the address of `marks[4]` if the base address = 1000. (5)

Que 8) Let `Age [5]` be an array of integers such that `Age [0] = 2`, `Age [1] = 5`, `Age [2] = 3`,

`Age [3] = 1`, `Age [4] = 7` Show the memory representation of the array and calculate its length. (5)

Que 9) Write a program to print the position of the smallest number of `n` numbers using arrays. (5)

Que 10) Write a program to read and display `n` numbers using an array. (5)

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Que 11) Data [] is an array that is declared as int Data [20]; and contains the following values:

Data [] = {12, 23, 34, 45, 56, 67, 78, 89, 90, 100};

- (a) Calculate the length of the array. (2)
- (b) Find the upper_bound and lower_bound. (2)
- (c) Show the memory representation of the array. (2)
- (d) If a new data element with the value 75 has to be inserted, find its position. (2)
- (e) Insert a new data element 75 and show the memory representation after the insertion. (2)

Que12) Data [] is an array that is declared as int Data [10]; and contains the following values:

Data [] = {12, 23, 34, 45, 56, 67, 78, 89, 90, 100};

- (a) If a data element with value 56 has to be deleted, find its position. (2)
- (b) Delete the data element 56 and show the memory representation after the deletion. (3)

Que 13) How do we Merge two unsorted Array? Show with an example. (3)

- b) How do we merge two sorted Array? Show the steps with an example. (7)

Ques 14) Explain 2-D Array. How do we declare Two-dimensional Array? (5)

Ques 15) Consider a 20 X 5 two-dimensional array marks which has its base address = 1000 and the size of an element = 2. Now compute the address of the element marks [18][4] assuming that the elements are stored in row major order. (5)

Ques 16) Write down any 5 differences between 1-D Array and 2-D Array. (5)